Geneva Ecola

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Education

Expected 2027 **Ph.D. Stanford University**, Electrical Engineering
Thesis: Harnessing Ambient Radio Signals for Ultra-Low Power Connectivity
Advisor: Zerina Kapetanovic, *Thesis Committee*: Howard Zebker, Joseph Kahn

2023 **M.S. Stanford University**, Electrical Engineering

2019 **B.S. Harvey Mudd College**, Engineering

Research Experience

2023 – Present **Stanford Smart Sensing Systems Lab**, Doctoral Researcher
Working on new ultra-low-power forms of communication including satellite backscatter connectivity and thermal noise backscatter communication.

Publications (*indicates co-first authorship)

2026 **First-Principles Channel Model and Capacity-Achieving Transmission Scheme in Thermal Noise Backscatter Links**
Geneva Ecola*, Jida Zhang*, Joesph Kahn and Zerina Kapetanovic
Proceedings of the National Academy of Science

2025 **Wireless Johnson Noise Thermometry for Passive Temperature Sensing**
Jasmin Falconer*, Geneva Ecola* and Zerina Kapetanovic
IEEE Journal of Microwaves

2025 **SARLink: Satellite Backscatter Connectivity using Synthetic Aperture Radar**
Geneva Ecola, Bill Yen, Ana Banzer Morgado, Bodhi Priyantha, Ranveer Chandra, and Zerina Kapetanovic
ACM SenSys

Professional Experience

Summer 2024 **Apple**, Radio Frequency Intern
Evaluated impact of chip packaging variants on radio receiver performance and automated testbed for optical chip-fiber alignment.

Summer 2023 **Tesla**, Wireless Hardware Intern
Evaluated localization technologies for a passive entry system including BLE and UWB. Designed and evaluated RF balun circuit matching between chip and antenna.

2019 – 2021 **Trellisware Technologies**, Senior Radio Frequency Engineer
Redesigned the high-frequency analog front-end of radios using simulations and tested hardware in lab. Led hardware sustainment of a software-defined radio digital board, coordinating team of 10+ engineers.

Summer 2018 **Echodyne**, Antenna Intern
Measured electronically scanning array antenna near field at multiple scan angles and frequencies and developed MATLAB scripts to quantify antenna performance metrics.

Fellowships

2026 **Stanford Dissertation Completion Fellowship**

2024 **Stanford Graduate Fellowship, Rambus Corporation Fellow**

2024 **Apple-Stanford Integrated Systems PhD Fellowship**
Proposal: Enabling Ultra-Low Power Smart Sensing Systems

2021 **Stanford Graduate Fellowship, Rambus Corporation Fellow**

Honors and Awards

2026 **Best Presentation at ACM MobiSys Rising Star Forum**

2026 **Best Poster Runner Up at ACM MobiSys N2Women Workshop**

2026 **Qualcomm Innovation Fellowship Finalist**
Proposal: Ultra-low Energy IoT Networks Using Thermal Noise Communications

2025 **Stanford Emerging Environmental Scholar Research Award**
Proposal: Bringing Smart Sensing Systems Everywhere on Earth

2023 **Stanford EE315 Best Analog to Digital Converter Design**

2019 **Harvey Mudd College Departmental Honors in Engineering**

2018 **Harvey Mudd College T. Larry Norin Memorial Scholarship**

2018 **Harvey Mudd College Best Engineering Clinic Presentation**

2015 **Harvey S. Mudd Merit Award**

Talks

2026 **Harnessing Ambient Radio Signals for Ultra Low Power Sensing and Communication Systems**
ACM MobiSys Rising Star Forum

2026 **Ultra-low Energy IoT Networks Using Thermal Noise Communications**
Apple Fellowship Talk

2025 **Doing More with Less: Harnessing Ambient Signals for Wireless Sensing and Communication**
University of Washington Department of Electrical Engineering
Apple Fellowship Talk

2025 **SARLink: Satellite Backscatter Connectivity using Synthetic Aperture Radar**
ACM SenSys
Apple Fellowship Talk

Posters

Ultra-low Energy IoT Networks Using Thermal Noise Communications
Geneva Ecola, Jida Zhang and Zerina Kapetanovic
2026 *ACM MobiSys N2Women Workshop*
2026 *Qualcomm Innovation Fellowship Finals*

Vision Assisted Beamsteering for Full Control of Wireless Power Transfer and Joint Communication Link
Jasmin Falconer, Geneva Ecola, and Zerina Kapetanovic
2025 *Stanford Center for Image Systems Engineering Affiliates Annual Meeting*

SARLink: Satellite Backscatter Connectivity using Synthetic Aperture Radar
Geneva Ecola, Bill Yen, Ana Banzer Morgado, Bodhi Priyantha, Ranveer Chandra, and Zerina Kapetanovic
2025 *ENSsys Workshop: Energy Harvesting & Energy-Neutral Sensing Systems*
2024 *Stanford Center for Image Systems Engineering Affiliates Annual Meeting*

Teaching

Coursework

2026 **Course Developer, Internet of Things, Stanford University**
Developed laboratory exercises and a final project that teach students to build connected devices using a range of wireless communication protocols.

2025 **Course Assistant, Circuits I, Stanford University**
Led laboratory sections and office hours for more than 30 students in an introductory circuits course. Delivered midterm and final review lectures.

2023 **Course Developer and Assistant, 3D+ Imaging Sensors, Stanford University**
Led laboratory sessions and office hours for more than 30 students on ultrasonic sensors, time-of-flight cameras, and mmWave radar. Designed a final project in which students developed a through-wall imaging system using mmWave radar.

2017 – 2018 **Teaching Assistant, Systems Engineering, Harvey Mudd College**
Taught foundational concepts for laboratory experiments and guided students through system debugging and troubleshooting.

Service

Professional Service

2026 – Present **Chair and Founder, Stanford IEEE Microwave Theory & Technology Branch**
Founded the student branch and organize technical workshops, invited talks, and networking opportunities to build a community of students interested in microwave and wireless.

2026 **Reviewer, IEEE Microwave Magazine**

2026 **Technical Program Committee Member, ACM MobiCom S3 Workshop**

2025, 2026 **Reviewer, IEEE Journal of Microwaves**

2025 **Poster Program Committee Member, ACM SenSys**

University Service

2023 – 2025 **Vice President, Stanford Women in Electrical Engineering**
Organized professional development, community-building, and networking events for an organization with more than 100 members.

2024 – 2025 **Student Life and Culture Committee, Stanford Electrical Engineering**
Represented graduate student perspectives in departmental initiatives related to student experience, community, and culture.

2022 – 2023 **Outreach Chair, Stanford Women in Electrical Engineering**
Organized a mentorship program for more than 50 participants and led hands-on fruit-battery demonstrations at Stanford community engagement events.